Analyze Power BI Dashboard Performance Using Performance Analyzer

1. Introduction

Performance Analyzer is a built-in feature in Microsoft Power BI Desktop that helps evaluate the loading and execution performance of dashboard visuals.

In this exercise, Performance Analyzer was used to measure the execution time of individual visuals in the Sales Performance Dashboard and identify components that may require performance optimization.

2. Objective

- Understand the Performance Analyzer feature in Power BI.
- Measure the loading time of individual dashboard visuals.
- Analyse DAX query execution and visual rendering time.
- Identify visuals with comparatively higher execution times.
- Understand the importance of dashboard performance optimization.

3. Tools and Technologies Used

- Microsoft Power BI Desktop
- Sales Performance Dashboard
- Performance Analyzer
- DAX (Data Analysis Expressions)

4. Implementation Steps

4.1 Opening the Optimize Tab

Opened the existing Sales Performance Dashboard in Power BI Desktop and navigated to the Optimize tab in the ribbon.

4.2 Opening Performance Analyzer

Selected Performance Analyzer from the Optimize tab. The Performance Analyzer pane appeared on the right side of the Power BI Desktop interface.

4.3 Starting Performance Recording

Clicked the Start Recording option to begin capturing the execution performance of the dashboard visuals.

4.4 Refreshing Dashboard Visuals

Selected Refresh Visuals to refresh all dashboard visuals and record their execution times.

4.5 Analysing Performance Results

Reviewed the recorded performance details for individual visuals.

The execution information included DAX query time, visual display time and other processing time.

4.6 Testing Dashboard Filters

Applied City and Date slicers to observe changes in visual performance under different filtering conditions.

5. Performance Analysis

The recorded execution times were reviewed to identify visuals that required comparatively more processing time.

The following components were considered:

- DAX Query: Time required to execute the DAX query.
- Visual Display: Time required to render the visual.
- Other: Additional processing time.

6. Performance Observations

The performance results were compared across different dashboard visuals to identify variations in execution time.

Visuals with comparatively higher execution times were identified for further investigation and possible optimization.

7. Performance Optimization Recommendations

The following practices can help improve Power BI dashboard performance:

1. Optimize complex DAX measures and calculations.
2. Avoid unnecessary columns and fields in the data model.
3. Reduce excessive visuals on a single report page.
4. Use appropriate data types and relationships.
5. Limit unnecessary data points in charts.
6. Prefer efficient data modelling and filtering techniques.

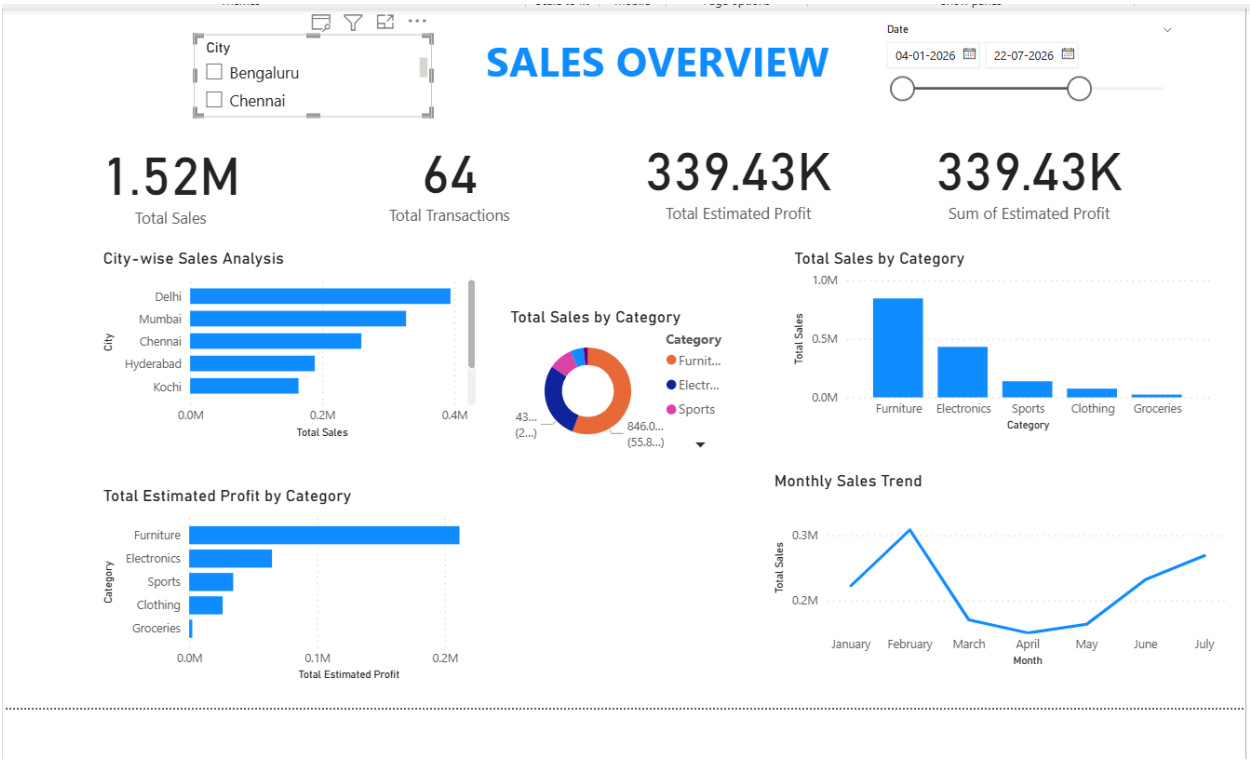
8. Conclusion

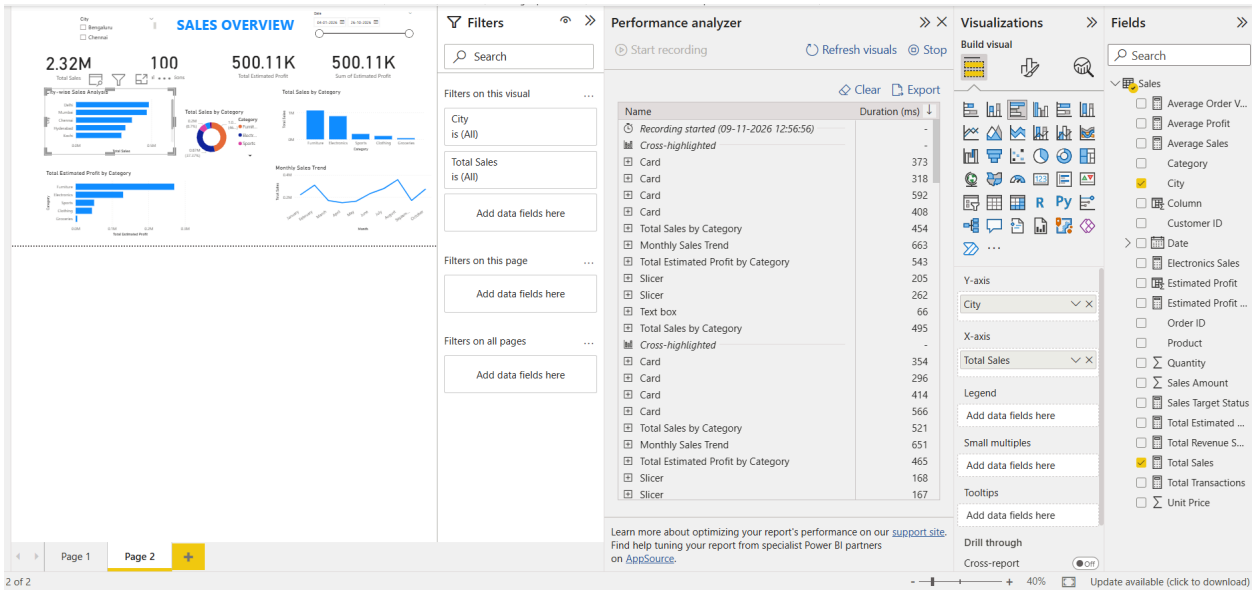
Successfully used Performance Analyzer in Power BI Desktop to evaluate the performance of the Sales Performance Dashboard.

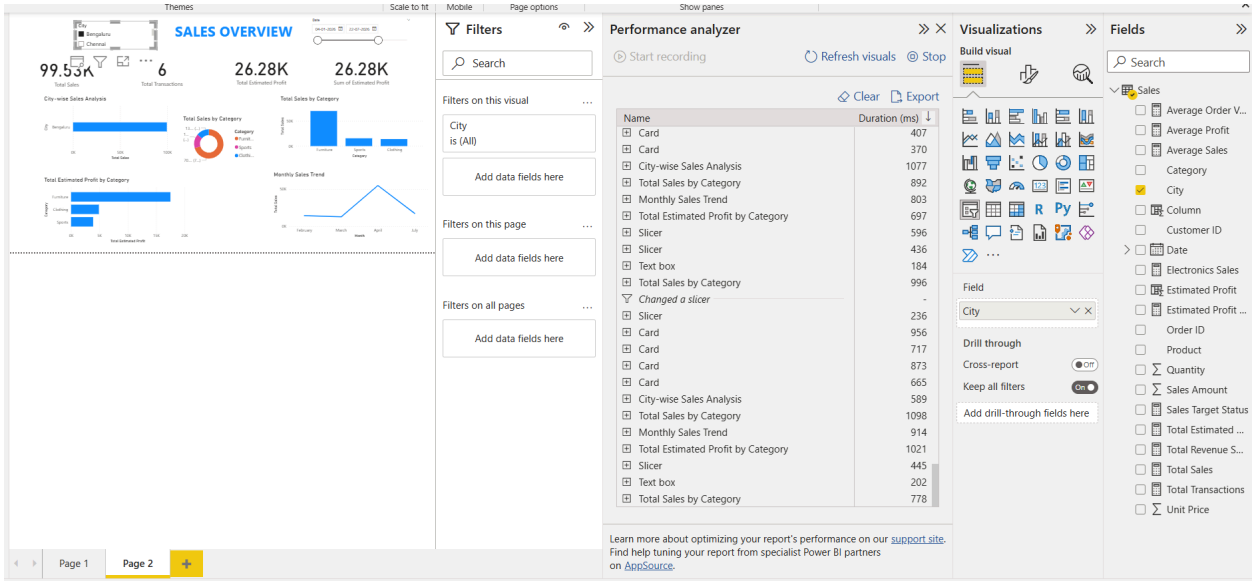
Recorded the execution time of individual visuals, reviewed DAX query and visual rendering times, and compared dashboard performance under different filtering conditions.

This exercise provided practical knowledge of Power BI performance monitoring and helped understand how Performance Analyzer can be used to identify potential dashboard optimization opportunities.

9. Screenshots and Evidence




SALES OVERVIEW
2.32M
100
500.11K
500.11K
Filters
Search
Filters on this visual
City
is (All)
Total Sales
is (All)
Add data fields here
Filters on this page
Add data fields here
Filters on all pages
Add data fields here
Performance analyzer
Start recording
Refresh visuals
Stop
Clear
Export
Name
Duration (ms)
Recording started (09-11-2026 12:56:56)
Cross-highlighted
Card 373
Card 318
Card 592
Card 408
Total Sales by Category 454
Monthly Sales Trend 663
Total Estimated Profit by Category 543
Slicer 205
Slicer 262
Text box 66
Total Sales by Category 495
Cross-highlighted
Card 354
Card 296
Card 414
Card 566
Total Sales by Category 521
Monthly Sales Trend 651
Total Estimated Profit by Category 465
Slicer 168
Slicer 167
Learn more about optimizing your report's performance on our support site.
Find help tuning your report from specialist Power BI partners on AppSource.
Visualizations
Build visual
Y-axis
City
X-axis
Total Sales
Legend
Add data fields here
Small multiples
Add data fields here
Tooltips
Add data fields here
Drill through
Cross-report
Off
Fields
Search
Sales
Average Order V...
Average Profit
Average Sales
Category
City
Column
Customer ID
Date
Electronics Sales
Estimated Profit
Estimated Profit ...
Order ID
Product
Quantity
Sales Amount
Sales Target Status
Total Estimated ...
Total Revenue S...
Total Sales
Total Transactions
Unit Price
Page 1
Page 2
2 of 2
40%
Update available (click to download)


Themes
Scale to fit
Mobile
Page options
Show panes
SALES OVERVIEW
99.53K
6
26.28K
26.28K
Filters
Search
Filters on this visual
City
is (All)
Add data fields here
Filters on this page
Add data fields here
Filters on all pages
Add data fields here
Performance analyzer
Start recording
Refresh visuals
Stop
Clear
Export
Name
Duration (ms)
Card 407
Card 370
City-wise Sales Analysis 1077
Total Sales by Category 892
Monthly Sales Trend 803
Total Estimated Profit by Category 697
Slicer 596
Slicer 436
Text box 184
Total Sales by Category 996
Changed a slicer
Slicer 236
Card 956
Card 717
Card 873
Card 665
City-wise Sales Analysis 589
Total Sales by Category 1098
Monthly Sales Trend 914
Total Estimated Profit by Category 1021
Slicer 445
Text box 202
Total Sales by Category 778
Learn more about optimizing your report's performance on our support site.
Find help tuning your report from specialist Power BI partners on AppSource.
Visualizations
Build visual
Field
City
Drill through
Cross-report
Off
Keep all filters
On
Add drill-through fields here
Fields
Search
Sales
Average Order V...
Average Profit
Average Sales
Category
City
Column
Customer ID
Date
Electronics Sales
Estimated Profit
Estimated Profit ...
Order ID
Product
Quantity
Sales Amount
Sales Target Status
Total Estimated ...
Total Revenue S...
Total Sales
Total Transactions
Unit Price
Page 1
Page 2